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Still in love with solar energy? Installation size, affect, and the social acceptance of renewable energy technologies

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ABSTRACT

Solar and wind energy are expected to play a key role in creating a climate-neutral Europe by 2050 and decarbonizing energy production in general, albeit requiring significant deployment. This presupposes that the population accepts such energy sources, thus necessitates understanding how people perceive energy systems. Unlike the abundant literature about wind energy, social acceptance of solar energy has received less attention, especially concerning large solar installations. Opinion polls indicate that solar energy enjoys a high level of socio-political acceptance and is preferred to other renewables, although it is unclear whether this acceptance persists as the technology is deployed on a large scale. This paper helps close this gap by describing the results of a representative survey (n = 601) conducted using a between-subject design to examine how attitudes of the public towards solar energy vary based on the size of installations, how the latter compare to attitudes towards wind energy, and what the role of affect is in the former. Results reveal that the stronger preference for solar power decreases to a similar level as that for wind energy when comparing installation of similar sizes, highlighting that solar energy installations may not easily be scaled up. The study also shows that affect plays an important role in forming people's attitudes towards wind and solar, especially concerning large-scale installations. This underlines the need for policymakers and project developers, who aim to deploy renewables on a large scale, to attend to the affective component of decision-making.

Author contribution

Julia Cousse: sole author.

1. Introduction

Recent research suggests that investment into climate-friendly policy initiatives may help the world move closer to a net-zero emissions pathway, as targeted in the 2020 European Green Deal [1], and could also supply the best economic returns on government spending [2]. Among the policies that are both most effective and sustainable are those that promote investment into greener infrastructure, in particular renewable energies, energy storage, and the modernization of the electricity grid [2]. Of all the renewables, solar energy is one of the fastest growing energy sources and could consequently contribute substantially to reducing dependence on fossil fuels and decreasing global CO_2 emissions [3]. However, apart from technical challenges, the social acceptance of renewables appears central to the latter. At first glance, the issue of the acceptance of renewables might seem unproblematic, as several studies have revealed a high level of general[1] acceptance of renewables [e.g. 4,5]. However, concerning the actual implementation of projects, policymakers face considerable local resistance [e.g. 6], which may increase exponentially as renewables are deployed on a larger scale. This is particularly problematic for a country such as Switzerland, which is lagging in terms of production of solar and wind energy and needs to increase its share of renewables significantly to meet the objective of reducing CO_2 emissions to zero by 2050 [7].

While many studies have focused on the social acceptance of wind energy in recent years [8–11], the social acceptance of solar energy, and more specifically of large solar photovoltaics, has received less attention [12–15]. However, while solar energy enjoys a high level of general acceptance of the population and appears to be greatly preferred to other renewable energy sources [4,16,17], it is yet unclear whether the preference for this technology persists if the technology is deployed on a larger scale. Accordingly, obtaining valid data about people's preferences for energy technologies, along with the factors that shape those preferences, can help policymakers overcome social-acceptance-related issues associated with the implementation of renewable energy projects

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[1] *General* acceptance of renewables is also referred to *socio-political* acceptance [27].

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[18,19].

Further, within the literature on the social acceptance of renewables, Devine-Wright [20] highlights that the implementation of energy projects can disturb emotional bonds to places. One might thus expect that such emotional bonds would be more strongly disrupted as the size of a technological installation increases. However, emotional reactions to the siting of solar energy projects are still underexplored, although they appear to be a key element in the creation of the attitudes of the population towards energy projects. An increasing body of literature has shown that affect – the specific quality of something being good or bad, experienced as a feeling [21] - plays an important role in decision-making [22–24], including in the context of preferences for energy sources [25,26].

Based on these considerations, it appears important to investigate whether the high acceptance level enjoyed by solar energy is likely to persist if the technology is deployed on a large scale,[2] or whether it may face resistance similar to that associated with wind energy projects. For this purpose, the current study investigates how specific attitudes[3] towards solar energy vary based on the size of the installation (solar rooftop versus solar park) and compare these to specific attitudes towards wind energy installations. Further, given the importance of affect in attitude formation, its explores how feelings elicited by solar and wind energy influence the relationship between energy technologies and attitudes. The specific research questions are presented below:

(1) How do differences in attitudes towards solar energy vary based on the size of installations, and how do the latter compare to attitudes towards wind energy installations?
(2) What is the role of affect in the development of attitudes towards the different energy installations?

In brief, the study finds that (1) attitudes towards solar and wind park installations of a comparable size do not differ; (2) affect has a mediating role in the formation of attitudes; and, (3) the affect elicited by energy installations have the largest effect on attitudes in the case of large-scale energy installations. The rest of the paper is organized as follows: Section 2 discusses and reviews relevant literature in the field of the social acceptance of renewable energies; Section 3 describes data and methods; and Section 4 presents the empirical findings. Finally, Section 5 discusses the most important findings in light of the literature, and includes a conclusion, policy implications, limitations, and suggestions for further research.

2. Literature review

2.1. Social acceptance of renewables

An increasing amount of research has investigated the social acceptance of renewable energy in recent years [e.g. 27,28]. Wüstenhagen et al. [27] list three different levels of social acceptance: socio-political, community and market acceptance. These three levels are not only interrelated but also involve different actors (project developers, policymakers, consumers) who are involved in different levels [29]. For example, an increase in challenges in the area of community acceptance (i.e., local acceptance) can impact pre-existing high levels of socio-political acceptance (i.e., general acceptance). This implies that social acceptance is a multidimensional and dynamic phenomenon [30,31].

Several studies have found high socio-political acceptance of the different types thereof [36,37]. In terms of the acceptance of renewable energy technologies, solar energy stands out due to its positive socio-political perception and acceptance [16,36,38]. However, high levels of socio-political acceptance are a necessary but not sufficient condition for the successful implementation of energy projects. Regarding wind energy, different studies have identified high levels of socio-political acceptance [31,39], but wind energy projects frequently face acceptance-related issues on a local level that impede their deployment [40]. Moreover, solar and wind energy have experienced impressive learning curves resulting in significant cost reductions [41,42] making renewables the most cost-effective source of new electricity generation [42]. This has led researchers to argue that community acceptance from the population is a main limiting factor in the successful implementation of renewable energy projects [10,11]. As such, an important question is to what extent high levels of socio-political acceptance translate into high levels of local acceptance of energy projects [16,27].

For many individuals, the transition to renewables is experienced as the extension of the energy system into places that were formerly untouched [43, p.335]. Such interference with previously "unaffected" land appears to play an important role in the acceptance of energy sources at the local level. Wolsink [44] has, for example, highlighted that the visual impact of wind turbines on the landscape plays a significant role in the overall acceptance of wind power. Research has also noted that decision-making about the spatial implications of the implementations of renewable energy projects mainly involves landscape issues, not only in terms of physical options for siting, but also the assessment of landscape values that may be affected by the infrastructure [45]. Regarding the latter, the concepts of 'visibility' and 'visual impact' must be distinguished [45] as the implementation of energy projects must be planned not only in terms of their visibility, but also in terms of their visual impact: namely, the change in the quality of the landscape induced by the siting of the energy infrastructure [45]. Changes in the quality of the landscape are subjective as they are mainly due to individuals' different assessments thereof. Accordingly, the relationship between visual impact, landscape and perception by the population is complex [45] and the implementation of projects that do not take into account how the visual impact of energy projects may disrupt people's values and emotions may hit roadblocks [20,45]. However, little research exists about the perception of the population concerning the visual impact of solar energy installations [e.g., 46] and, more importantly, how such perceptions may vary based on the size of the installation. Controlling for the size of the installation in the case of solar energy appears to be particularly relevant, as previous research that used a Swiss sample, revealed that people associate solar power specifically with the placement of solar panels on rooftops, and not large-scale installations [16]. It is thus yet unclear whether the positive symbolic value attached to solar energy and in turn, high acceptance levels, may be sustained for large-scale solar power plants [16].

While some research has measured socio-political and community acceptance of solar energy [36,47], most has focused on wind energy, with the exception of a few cases. Vuichard et al. [47], using a Swiss sample, found that a majority of respondents would be in favor of solar parks in an alpine region, and that local ownership increases the level of acceptance. Wissen et al. [15], also using a Swiss sample, showed that deployments of rooftop PV installations are preferred to wind energy and open-space PV installations. Moreover, Carlisle et al. [14] found, using a US sample, that the public broadly supports the deployment of large-scale solar projects, and this support is not different between deployment considered in general or when located in one's own county. While these studies offer useful insights, it is still unclear if the preference for solar energy over wind energy would be sustained if deployments of a similar size were compared, especially in a smaller country such as Switzerland.

Previous research into the social acceptance of wind energy provides

[2] Note: *Scale,* in the current study refers to the size of the energy infrastructure only [e.g., 32, 33] as opposed to local and national levels, for example [e.g., 34].

[3] "Specific attitude" refers to attitudes towards local energy projects, "e.g., what people think about wind energy projects in their vicinity, regardless of the specific project characteristics [35, p.79]".

important information on which to base an analysis of attitudes towards solar energy, and specifically, about how attitudes may change based on the size of installations. Factors related to the proposed size of a project have been shown to be significantly related to their support, with larger energy projects being less likely to be favorably received [44,48,49]. Further, Devine-Wright and Batel [50] highlighted the need to reconsider the "localist" perspective adopted by earlier research. Specifically, the authors noted that the literature on the social acceptance of renewables can be criticized for taking on a narrow spatial focus. They argued that low carbon energy infrastructure does not only involve local projects. Instead, while having an evident local impact, it also involves concerns on multiple levels [43,50,51]. For example, people may feel loosely attached or estranged from the place in which they presently live [50,52,53]. In contrast, people may feel a strong attachment to localities other than those in which they now live, such as places where they have previously lived [49,54,55]. To contribute to this relatively new way of assessing attitudes towards local energy projects, the current study takes into account these aspects in how it measures the related attitudes by not only assessing attitudes about "local projects" involving energy projects close to respondents' homes, but also in other places that individuals may feel close to.

2.2. Affect and the social acceptance of energy sources

People's judgments and decisions are oftentimes based on their feelings about objects (e.g., wind turbines) [56]. Feelings, which are part of the affective system of decision-making, are based on associative and intuitive processes that are generally faster than cognitive processing [57]. The affect elicited by an object can serve as a quick and efficient signal that contributes to decision-making [58] given that positive or negative feelings associated with objects consequently guide their evaluation [59]. However, energy-related preferences have mainly been investigated from a cognitive perspective [e.g.60–62], and only a few studies have examined affective factors [16,26,63,64].

Researchers in the field of psychology have used word-association techniques for a long time [65–68]. Affective imagery, which is employed in the current research to measure affective reactions to solar and wind energy, uses word association to reveal the positive and negative associations elicited by a stimulus [21]. Researchers have used affective imagery to investigate different domains, including GM food [68], nuclear power [22], and smoking addiction [69].

Specific to energy sources, Jobin and Siegrist [38] highlighted that affect guided preferences for energy portfolios, and that solar energy generally elicited more positive affect than wind energy. Further, Truelove [26] found that affect was significant in predicting attitudes towards energy project implementation. Based on the former, the hypothesis is that feelings may have a mediating effect on the relationship between an energy installation and attitudes towards it. To the best of the author's knowledge, although previous findings have highlighted that affect is important in the formation of attitudes towards energy source infrastructures, no studies have yet investigated how affective reactions to solar energy can vary based on the size of installations, or how the affect elicited by solar energy infrastructures of different sizes compares to the affect elicited by wind energy infrastructure.

3. Data and methods

3.1. Sample

Data were collected through a representative survey of 601 Swiss respondents between March and April 2019. The respondents of the study were selected through a Swiss online panel (n = 100,000).[4] The heart of the questionnaire was a between-subject experiment about the affect and attitudes evoked by solar and wind energy installations. Data about further variables were also collected: general affective reactions to solar and wind energy, gender, age, language region, political orientation, place of residence, education level, and income. Details about each of the variables can be found in Table A, Appendix A. The completion rate was 81.65%.[5] The sample was representative of the Swiss population with regards to gender, age, language region, and political orientation (see Table 1). In addition, 19.0% of respondents indicated living in a city center, 12.6% in the suburbs, 20.1% in an agglomeration, and 47.4% in rural areas, controlling for a potential urban versus rural divide. The Italian and Rhaeto-Romanic speaking regions of Switzerland were not included in the sample, as they constitute only 8% and 0.5% of the Swiss population.

3.2. Survey design

All participants (n = 601) were first surveyed to reveal their general spontaneous associations and related feelings (or affect) elicited by the stimulus wind and solar energy using the method of continued word association [22,72]. In a second step, an experimental between-subject design was implemented, and participants were presented with pictures of a solar park, wind park, or rooftop installation within the same landscape (Figure B, Appendix B). In the experimental between-subject design, each participant was randomly assigned to one treatment group and presented with pictures of a solar park (Group 1, n = 197), rooftop solar (Group 2, n = 203), or wind park (Group 3, n = 201). The use of pictures was deemed to be particularly relevant in the context of this study to ensure that participants mentally visualized similar installations. Additionally, the use of pictures in surveys has shown to be

Table 1
Structure of sample. Data sources: Gender [70], Age [70], Political orientation [70], and Language region [71].

	Total sample	Swiss population
	n = 601	
Age		
15–29	18.50%	23%
30–39	14.50%	18%
40–49	20.00%	19%
50–59	18.80%	20%
60–74	28.30%	20%
Gender		
♂	50.10%	49%
♀	49.90%	51%
Education		
low/medium	63.10%	62%
high	36.90%	38%
Political Orientation		
Conservative (SVP, FDP, CVP)	57.40%	57%
Center-progressive (BDP, GLP)	8.00%	9%
Left-wing (SP, GPS)	26.50%	26%
Other/none	8.20%	8%
Language region		
French-speaking	34.10%	27%
German-speaking	65.90%	73%

[4] The panel was managed by Intervista's. Their panel approach helps create a sample almost equivalent to a probability sample of the Swiss voting population (https://www.intervista.ch/?lang=en) as opposed to opt-in panels.

[5] This figure describes the proportion of completely finished interviews in relation to the number of individuals who clicked on the link to the survey.

very helpful for understanding people's perceptions [12,73–75]. The method of continued word associations was also used to measure affective reactions to the pictures. In total, respondents were exposed to two pictures in two very different settings (referred to as Scenario 1: Alpine, and Scenario 2: Midlands) to control for the effect of the surrounding landscape and test whether differences observed between solar and wind energy installations were sustained in different settings. This is important, as past studies have highlighted that perceptions of renewable energy infrastructures such as wind turbines or photovoltaic systems not only depend on the infrastructure itself, but also on the visual impact of the energy infrastructures. The latter involves not only an assessment of the infrastructure as such, but of landscape quality change induced by the siting of the latter [12,45,76]. The pictures were selected to represent environments, which were not sterile but instead could elicit affective reactions from respondents (i.e., real Swiss landscapes). Further, the two pictures were presented in a random order to control for contamination effects. Note that the design of the energy infrastructure presented in the pictures does not account for regulations relating to whether an energy project could legally be implemented in the given setting, although regulatory issues play an important role in the selection of site development. At the end of the survey, each group was surveyed about their attitude towards the implementation of a hypothetical solar park, solar rooftop, or wind park installation. Sociodemographic data about each subgroup can be found in Table C, Appendix C.

3.3. Measures

3.3.1. Affective reactions

Affective reactions were measured following the method of continued word association, as described by Peters and Slovic [22]. To measure the general spontaneous associations and related affective evaluations elicited by wind and solar energy, respondents were asked "What are the first thoughts or images that come to your mind when thinking of solar/wind energy?".[6] In a second step, each subgroup was asked: "What are the first thoughts or images that come to your mind when looking at this picture?" (installation-specific level). Respondents were asked to give a minimum of one association, and a maximum of five. Each association provided by the respondents was subsequently evaluated by them on a seven-point affect scale from 1 ("very negative") to 7 ("very positive"). If a respondent offered five responses, the affective evaluations of each of these was averaged to develop the mean affective evaluation. In the analysis, the mean affective evaluation of the associations was used, which corresponded to the average of every affective evaluation of each respondent. The descriptive statistics of the variables can be found in Table D, Appendix D.

3.3.2. Local concern

To measure attitudes towards the implementation of the energy infrastructure, a proxy, local concern, which is an attitudinal measure, was used. Further, attitudes to the implementation of the energy infrastructure close to respondents' homes or other places they may feel close to, rather than attitudes towards the specific projects depicted in the pictures were measured. Specifically, local concern was measured by asking respondents to specify their level of agreement with the following statement: "I would be concerned if a wind park/solar park/solar rooftop installation was built close to my neighborhood or in other places I feel close to," on a scale from 1 ("totally disagree") to 5 ("totally agree"). The design of this question – and in particular, the addition of the phrase "(…) in other places I feel close to" – was motivated by recent research by Devine-Wright and Batel [50], which highlighted the need to reconsider the "localist" perspective adopted by previous research. An option to select "I do not know" was also available. Respondents were excluded from the analysis if they selected this option (n = 59).

4. Results

4.1. Attitudes towards solar park-, rooftop solar-, and wind park installations

Fig. 1 shows respondents' average level of local concern about the wind park (M = 3.16, SE = 0.11, N = 201), solar park (M = 2.89, SE = 0.11, N = 197), and rooftop solar installations (M = 1.67, SE = 0.08, N = 203). As noted in the methodological section of the paper, local concern measured attitudes towards hypothetical projects close to respondent' homes or other places they may feel close to, not attitudes towards the projects in the pictures. Based on a one-way Welch ANOVA[7] and Games-Howell post-hoc tests ($F = 76.49$, $df_1 = 2$, $df_2 = 389.87$), the results indicate that the level of concern about the solar park and the wind park installation ($p = 0.16$) was not significantly different. In contrast, the level of concern about solar rooftop installations was found to be significantly different from that expressed in relation to the solar park ($p < 0.001$) and wind park installations ($p < 0.001$).

4.2. Affective reactions elicited by wind and solar energy

To investigate how affect varied between wind and solar energy, an investigation of the feelings or affect elicited by wind and solar energy in general, meaning the affect elicited by the terms "solar energy" and "wind energy" was first undertaken, and the distributions of the affect evoked among the overall sample were evaluated using box plots (n = 601) (Fig. 2).

The distributions of the feelings elicited were not uniform for both solar and wind energy. Over 80% of respondents (83.72%) showed positive affect towards solar energy in general (Median = 6, IQR = 2), whereas only slightly fewer than 60% (57.81%) of respondents demonstrated positive affect towards wind energy in general (Median = 4.75, IQR = 3). The two boxplots further highlight that there exists more variability in affect in the case of wind (SD = 1.83) than in the case of solar (SD = 1.45). Conducting a one-way Welch ANOVA, it was found that the difference in affect between solar and wind, measured at a general level, was significant (N = 1204, F = 469.52, df = 1, $p < 0.001$), being more positive in the case of solar power (Mean = 5.74) than in the

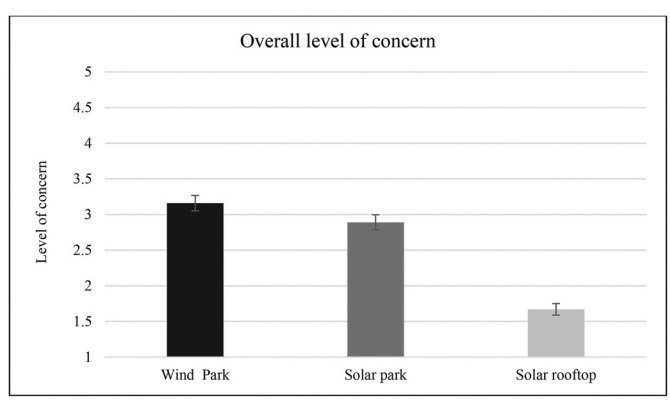

Fig. 1. Overall level of local concern for the three different experimental groups of respondents.

[6] This question was asked in the exact same form in Cousse et al. [64].

[7] A Levene's test was significant at $p < 0.001$. A Welch's ANOVA test was conducted ($F = 76.49$, $df_1 = 2$, $df_2 = 389.87$, $p < 0.001$) and was significant.

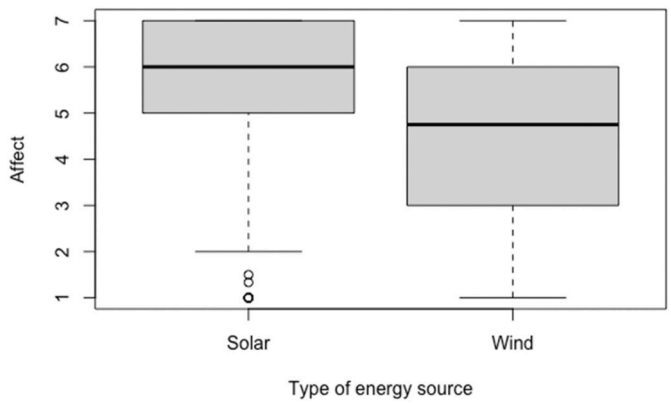

Fig. 2. Boxplots of affect elicited by wind and solar energy at a general level, displaying the median, 25th and 75th percentiles.

case of wind power (Mean = 4.49). These results suggest that when individuals think about the two energy sources on an abstract level, solar energy elicits more positive feelings than wind energy.

4.3. Affective reactions elicited by wind and solar energy installations

Next, the affect elicited by the solar and wind energy installations, including differences between solar rooftop and solar park installations was investigated. The results were more nuanced than those found at a more general level. Fig. 3 shows the affect elicited by solar and wind installations in Scenario 1 (Alpine). Affect here is not only elicited by the energy installations, but also by other elements in the pictures. As such, spontaneous associations and the related affect cannot be solely attributed to the energy installation, but also to other stimuli present in the pictures. Nonetheless, the comparison of energy installations in the exact same landscape controls for noise (i.e. other elements in the picture) and permits the investigation of differences in affective reactions in relation to the installations themselves.

The greatest standard deviation was found in the case of the affect elicited by wind parks (N = 201; SD = 2.18), followed by solar parks (N = 197; SD = 2.06), and then solar rooftops (N = 203; SD = 1.04). The

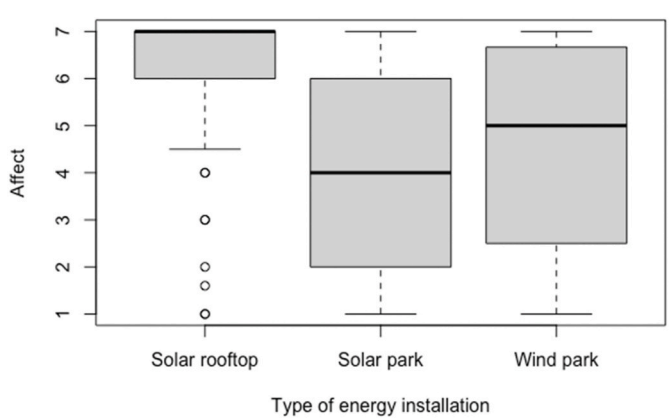

Fig. 3. Boxplots of affect elicited by wind and solar energy in Scenario 1 (Alpine), displaying the median, 25th and 75th percentiles.

solar rooftop installations (Mdn = 7, IQR = 1) elicited positive affect among 96.06% of the sample, while affect appears to be less homogeneous in the case of the wind park or solar park installations. For example, for the solar park installation (Mdn = 4, IQR = 4), 47.33% of respondents showed generally positive affect towards it. After conducting a one-way Welch ANOVA with Games-Howell post hoc tests,[8] it was further found that the affective reactions elicited by the wind park installation (Mdn = 5, IQR = 4.42, M = 4.50) were significantly different from those elicited by solar rooftop installations (M = 6.43, $p < 0.001$), but not from those elicited by solar park installations (M = 4.05, p = 0.09). Independent t-tests further highlighted that affective reactions in this scenario are significantly different between the solar park and solar rooftop installations (t = −14.69, $p < 0.001$).

Fig. 4 shows the results of the affective reactions elicited in Scenario 2 (Midlands). The distributions of the affect elicited by wind park- (SD = 1.92) and solar park (SD = 1.92) installations are associated with similar standard deviations. It was further observed that for the solar park installation (Mdn = 4.2, IQR = 3), 50.25% of respondents showed overall positive affect. In contrast, 68.84% showed positive affect towards the rooftop installation in the same scenario (Mdn = 5.2, IQR = 2.5). Further, a one-way Welch ANOVA with Games-Howell post hoc tests[9] showed that affective reactions elicited by the wind park installation (Mdn = 4, IQR = 3.5, M = 3.89) are significantly different from those elicited by the solar rooftop installations (M = 4.99, $p < 0.001$), but not from those for solar park installations (M = 4.32, p = 0.07), similarly to the findings for Scenario 1 (Alpine). Independent t-tests further highlighted that affective reactions in this scenario are significantly different between the solar park and solar rooftop installations (t = −3.70, $p < 0.001$).

4.4. The mediating role of affect in attitudes towards wind and solar energy infrastructures

Model 4 of the PROCESS macro for SPSS was used to investigate the role of affect in the formation of attitudes towards the energy installations. Separate multiple regression analyses for each energy source and installation were also undertaken. This was done to measure the distinct effect of affect on attitudes towards the energy infrastructure, as well as how this may differ from the general affect elicited by the energy source (solar versus wind energy) and the affect elicited by the installations while controlling for demographic variables.

To measure the potential mediating effect of affect, the variables

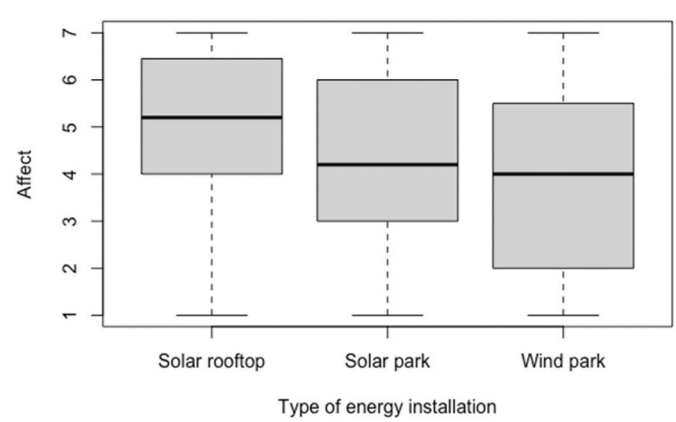

Fig. 4. Boxplots of affect elicited by wind and solar energy in Scenario 2 (Midlands), displaying the median, 25th and 75th percentiles.

[8] A Levene's test was significant at $p < 0.001$. A Welch's ANOVA test was conducted (F = 143.46, $df_1 = 2$, $df_2 = 350.02$, $p < 0.001$) and was significant.

[9] Levene's test was significant at p = 0.02. Welch's ANOVA (F = 19.36, $df_1 = 2$, $df_2 = 396.09$, $p < 0.001$).

measured were first defined in the following way: attitudes towards the energy technologies as the dependent variable, the type of energy installation (1 = wind park, 2 = solar park, 3 = solar rooftop) as the independent variable, and affect elicited by the pictures as the mediating variable[10]. In addition, the following variables were included as control variables: gender, age, political orientation, level of education, and place of residence (rural versus urban).[11] Bias-corrected 95% bootstrap confidence intervals were subsequently used to evaluate the indirect effect of the type of energy installation on attitudes, which was found to be below zero, at −0.30 (−0.41 to −0.24). The type of energy installation had a direct effect on the affect elicited, which, in turn, had an effect on attitudes. Therefore, a mediating effect of affect on the relationship between the type of energy installation and attitudes was found (see Fig. 5).

An investigation of how affect and attitude may vary based on the size of the energy installation was also performed. Attitudes towards the solar rooftop and solar park installation were defined as dependent variables, type of installation (solar rooftop versus solar park) as independent variables, and affect elicited by the energy technologies as the mediating variable. The same control variables were used as above. The confidence intervals used to evaluate the indirect effect of the size of the energy installation on attitudes was found to be below zero, at −0.64 (−0.84 to −0.46). The size of the energy installation had a direct effect on affect elicited, which in turn, had an effect on attitudes. Therefore, a mediating effect of affect on the direct relationship between installation size and attitudes was found (see Fig. 6).

Tables 2 and 3 present the results of the regression analyses. In Table 2, the effect of *general affect or feelings* on attitudes towards a wind park, solar park, or rooftop solar installation is highlighted. Affect was found to be significant, as well as the most important predictor of attitudes in all the regression models. The more negative the affect elicited, the higher the level of local concern. Specific to Models 1a to 1c, for which affect was measured at a general level, results show that affect had the highest level of significance in the case of attitudes towards a wind park ($B = -0.28$, $p < 0.001$) and the lowest in the case of attitudes towards a solar park ($B = -0.29$, $p = 0.004$). Overall, Model 1a, which predicts attitudes towards a wind park, explained 15.6% of the variance (adjusted R^2), 5% in the case of a solar park, and 12.4% in the case of a solar rooftop installation. Moreover, comparison of Models 1b and 1c shows that the affect elicited by solar energy at a general level seems to have had a greater influence on attitudes towards solar rooftop installations ($B = -0.18$, $p < 0.001$) than solar park installations ($B = -0.26$, $p = 0.004$). Multiple regressions controlling for income were also run. The results are reported in Appendix E, in Table E.1 and Table E.2. As with most surveys, about 15% of all observations are lost by controlling for income as respondents are often reluctant to report this. The robustness analysis shows that controlling for income has the following consequences on the model: First, in the case of Models 1b and 1c, general affect decreases in significance. Second, for Model 1b, the overall model becomes non-significant, while in Model 1c, the overall model is still significant but at a lower level. Overall, however, adding income does not impact the significant impact of general affect on attitudes.

Investigating the effect size (η_p^2) of affect[12] elicited by solar and wind energy in general on attitudes, it was further found that the former is greatest in the case of the wind park installation ($\eta_p^2 = 0.14$), and that the effect size is also larger in the case of the rooftop solar installation ($\eta_p^2 = 0.08$) than the solar park installation ($\eta_p^2 = 0.02$).[13]

Table 3 presents the impact of affect elicited by each of the energy installations presented in the pictures, controlling for the demographic variables. In general, Models 2a-2c explain a higher share of the former than Models 1a-1c. This improvement may indicate that the closer the stimulus is to the object under consideration (i.e., the energy infrastructure), the more powerful affect is at predicting attitudes. The robustness analysis, including income, shows that the effect of income had no influence on the significance of affect on attitudes (see Appendix E).

Further, using eta squared (η_p^2), a large effect size in the case of the wind park ($\eta_p^2 = 0.2$) and the solar park ($\eta_p^2 = 0.16$) installations, and a medium effect size in the case of the solar rooftop installation ($\eta_p^2 = 0.13$) were found.

Moreover, in some instances, the sociodemographic variables were significant, but not as important in terms of predicting attitudes as affect. Interestingly, whether respondents were French- (=1) or German-speaking (=0) mattered in relation to attitudes towards wind park and rooftop solar installations with French-speaking respondents having a more negative attitude towards wind park installations than German-speaking ones, but a more positive one towards solar rooftop installations. Political orientation (1 = conservative, 2 = center-progressive, 3 = left-wing) had a significant effect in the case of rooftop solar installations, with respondents from more conservative political parties being more concerned. This contrasts with attitudes towards wind park and solar park installations, for which political orientation was not significant at all. Further, adding income, as reported in Appendix E, it was found that the significance of political orientation became less significant in the case of Model 1c, and was no longer significant in the case of Model 2c in terms of predicting attitudes towards solar rooftop installations.

5. Discussion

As different countries are revising their energy strategies with the aim of decarbonizing the energy system, the acceptance of renewables at the community (local) level will be a key element in the success of this energy transition and proposed energy policies. Earlier literature that compared preferences for different renewable energy sources did not control for the size of installations when determining respondent preferences. However, lacking this approach, it is difficult to gauge which types of installations respondents were expressing their preferences for. Installation size appears to be critical, as it may amplify the visual impact of an energy installation on the landscape (namely, the change in landscape quality caused by the siting of the infrastructure) - which has been shown to be a dominant factor in the case of wind energy [45,76]. To address this gap and contribute to the research stream about the social acceptance of renewable energy and more specifically how the generally high acceptance level (i.e., socio-political acceptance) of renewables may (may not) translate into a high acceptance level at the community level, the present study investigated how attitudes towards solar energy compare to attitudes towards wind energy, while controlling for the size of the installation. Further, to better understand what shapes differences in attitudes, the research described in this paper

[10] In this case, the average of the affect elicited in Scenario 1 and Scenario 2 was used to create a single measure for affect.

[11] The sample size for the analyses of Section 4.4 is n = 548. Forty-nine respondents were excluded as they did not indicate allegiance to any specific party, or indicated a party other than one of the main seven Swiss political parties. Further, five respondents did not indicate whether they lived in a rural or urban area (one respondent did not have any allegiance to any specific party and did not indicate whether he lived in a rural or urban area).

[12] In this case, affect was operationalized in the following way: 0 = negative – for scores of 3 or lower; 1 = neutral – for scores of 4; 2 = positive – for scores of 5 or higher on the 7-point affective scale described in the methodology section.

[13] According to Ellis [77] and Cohen [78], a partial eta square (η_p^2) of 0.01 corresponds to a small effect size, 0.06 to a medium effect size, and 0.14 to a large one.

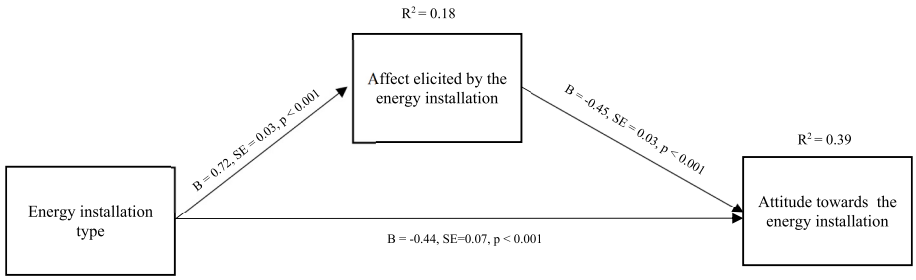

Fig. 5. Affect elicited by the energy installation mediate the relationship between the energy installation type and attitude towards it.

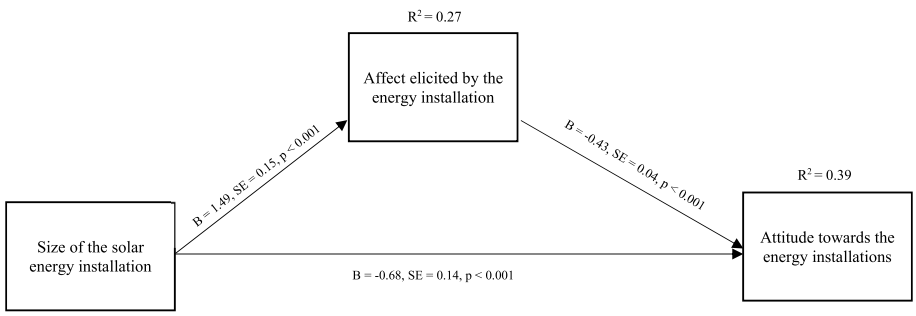

Fig. 6. Affect elicited by the size of the energy installation mediate the relationship between the size of the energy installation and attitude towards it.

Table 2
Multiple regression results concerning attitudes with general affect.

	Wind park (Model 1a)				Solar park (Model 1b)				Solar rooftop (Model 1c)			
	B	SE	β	p-values	B	SE	β	p-values	B	SE	β	p-values
General affect	−0.28	0.06	−0.35	0.00	−0.26	0.00	−0.22	0.00	−0.18	0.05	−0.25	0.00
Gender	−0.01	0.21	−0.00	0.96	0.18	0.23	0.06	0.45	0.05	0.16	0.02	0.76
Age	0.03	0.07	0.03	0.67	0.08	0.08	0.07	0.32	−0.00	0.05	−0.00	0.98
Language region	0.65	0.22	0.21	0.00	−0.18	0.24	−0.06	0.45	−0.45	0.17	−0.19	0.01
Education	−0.21	0.22	−0.07	0.33	0.29	0.24	0.09	0.23	−0.15	0.17	−0.07	0.37
Political orientation	−0.09	0.12	−0.05	0.46	−0.06	0.12	−0.04	0.62	−0.21	0.09	−0.16	0.03
Place of residence	0.02	0.09	0.02	0.80	0.17	0.10	0.12	0.11	−0.04	0.06	−0.04	0.54
R^2 adjusted	0.16				0.05				0.12			
F	5.85 (7, 177) ***				2.33 (7, 170)*				4.71 (7, 177)***			
Observations	185				178				185			

Note: for each type of energy infrastructure, a distinct multiple regression analysis was ran. B is the non-standardized regression coefficient. β is the standardized regression coefficient. SE is the standard error. General affect is specific to each energy source (i.e. wind and solar).

investigated the role of affect, a better understanding of which may help move beyond treating opponents of renewable installations as irrational individuals [29,79] by increasing understanding of their emotions.

5.1. Attitudes towards solar and wind energy installations: solar energy is not always preferred

A first novel finding pertains to attitudes towards solar and wind energy infrastructure when controlling for the size of the installations. Results showed no statistically significant differences in attitudes towards solar and wind park infrastructures. On the other hand, individuals appeared significantly less concerned about the implementation of solar rooftop installations. This result indicates that comparing attitudes towards different technologies but not taking into account the size of infrastructure may reveal preferences that do not endure as the related technologies are deployed on a larger scale. This finding is important, as European energy policies have established clear targets for renewables, but the configuration of their deployment is not yet clear. The latter factor is particularly important as the transition from fossil fuels to renewables has resulted in the proliferation of new types of landscapes that are sometimes contested, spurring interest in the relation between land use and social acceptance [80]. In this regard,

Table 3
Multiple regression results concerning attitudes with the affect elicited by the energy installations.

	Wind park (Model 2a)				Solar park (Model 2b)				Solar rooftop (Model 2c)			
	B	SE	β	p-values	B	SE	β	p-values	B	SE	β	p-values
Constant	5.20	0.51		0.00	4.87	0.57		0.00	4.27	0.52		0.00
Affect – energy installation	−0.43	0.05	−0.52	0.00	−0.45	0.06	−0.53	0.00	−0.36	0.07	−0.36	0.00
Gender	−0.14	0.19	−0.05	0.43	0.24	0.21	0.08	0.25	0.05	0.15	0.02	0.79
Age	−0.04	0.07	−0.04	0.55	−0.08	0.07	−0.08	0.24	−0.04	0.05	−0.05	0.46
Language	0.59	0.20	0.19	0.00	0.16	0.22	0.05	0.47	−0.44	0.17	−0.18	0.01
Education	−0.09	0.20	−0.03	0.64	0.17	0.21	0.05	0.42	−0.17	0.16	−0.07	0.29
Political orientation	−0.12	0.11	−0.07	0.28	−0.13	0.11	−0.08	0.23	−0.19	0.09	−0.15	0.03
Place of residence	0.01	0.08	0.00	0.95	0.08	0.09	0.06	0.38	−0.00	0.06	0	1
R^2 adjusted	0.29				0.28				0.18			
F	11.58*** (7, 177)				16.40*** (7, 170)				6.88*** (7, 177)			
Observations	185				178				185			

Note: Affect is specific to each type of energy infrastructure.

Wolsink [45] argues that local people and politics, not project developers, should take center stage in decisions about site selection and the use of land. The latter may be of particular relevance because renewable infrastructure is increasingly being deployed, and at a larger size, meaning that its visual impact is becoming more significant. Correspondingly, while in the case of Switzerland the population voted in favor of significantly increasing the share of national renewables through the Energy Strategy 2050, there is still hardly any awareness of what the implementation of this strategy means in spatial terms [15], while the active involvement of the latter in site selection may be critical in terms of the acceptance of the former [45].

In sum, policymakers should be aware that despite the general high acceptance of renewables, there exists the risk that any technology, if deployed on a large scale, might face resistance. Specific to solar energy, solar parks may face resistance connected to land availability and use of land and to their visual impact - factors that have already been identified in the case of wind parks [45,81]. On the other hand, solar rooftop systems do not occupy productive land [81]. However, researchers have pointed out the crucial role of large-scale solar installations for achieving the European target of climate neutrality by 2050 [82], which in fact substantiates the importance of further investigating the social acceptance of large-scale solar installations and the related acceptance of the necessary land use in comparison to alternative uses, in addition to ownership structure, which has already been identified as a highly relevant factor [29,45,47,83].

5.2. Solar energy: love decreases as the size of installations increases

Further, by comparing affective reactions to solar and wind energy on different levels (general vs. installation), and based on two different sizes of solar energy installation, the study produced novel findings in relation to the difference in perceptions about the two technologies. Specifically, when individuals think about solar and wind energy in general, with no further details provided about the two energy sources or the presentation of a specific project, the difference in affective reactions appear to reflect the differences in public attitudes observed in national polls, in which solar energy is greatly preferred over wind [36]. However, when affect is measured in relation to specific items of energy infrastructure presented in identical settings, new relationships emerge. While affective reactions to solar rooftop installations are significantly more positive than for those to wind and solar park installations, affective reactions in relation to solar park and wind park installations are similar, and these results hold true with different background settings. As a result, the study highlights that affective reactions are not related to energy sources per se, but rather to infrastructure as well as the scale of the installation, and thus provide a more nuanced picture about the previously identified "unanimously positive imagery of solar power" [16, p.362].

5.3. The important role of affect in attitude formation regarding energy infrastructure

Another important element that arises from the analysis is that affect matters in developing attitudes towards an energy project. Affect was found to have a mediating role in forming attitudes towards energy installations. Further, affect was the most relevant factor underlying attitudes towards a wind park, solar park, and solar rooftop installation. Specifically, the more positive the affect, the lower the level of concern about the implementation of the energy project.

Also interesting was the fact that when measured at a general level, affect had a larger effect on attitudes towards the implementation of a wind park than it did in relation to a solar park or solar rooftop installation. Also, affect elicited by solar energy in general appeared to be a better predictor of the acceptance of solar rooftop installations than solar parks. The conclusion is that individuals typically thought of solar rooftop installations when thinking about solar energy at a general level.

Further, the results highlighted that the regression models that included affect elicited by the solar and wind energy installations compared to solar and wind energy in general had greater explanatory power in terms of attitudes. This suggests that the more concrete an energy project appears to an individual, the more affect seems to matter in attitude formation. Additionally, the different sizes of the effect of affect on attitudes highlighted that the latter may matter more in the case of large-scale than small-scale installations. In the introduction, it was noted that Devine-Wright [20] highlighted that the implementation of energy technologies may disturb emotional bonds to a place, and that we can expect that these bonds will be more strongly disrupted as the size of the energy infrastructure increases. The results seem to support this assumption. Debates concerning the implementation of energy projects may thus become increasingly emotionally loaded as the size of

projects and the general deployment of renewable energy projects increases. In general, in line with earlier work [32–34], the current study highlights a negative relationship between project size and public support. Future research may investigate whether many small projects rather than a few large ones may be preferred by the population.

Finally, results show that French-speaking respondents are more concerned about the implementation of a wind park than German-speaking respondents, and the opposite holds true in the case of a solar rooftop installation. With more wind parks being planned in the French-than in the German-speaking part of Switzerland, a larger part of the French-speaking population may be in the second phase of the U-shaped curve, the phase during which the population is most critical. Interestingly, however, no differences between the two language regions in the case of attitudes towards solar parks were identified. This may be caused by the low level of familiarity or experience of the Swiss population with solar parks.

5.4. Limitations and future research

As with any research paper, there are some limitations that can the source for future research. First, the study focused on a specific geographical context. Switzerland is similar in many regards to other countries that have become engaged in the low-carbon energy transition, yet it also has specific characteristics that may affect the research findings, such as a relatively high per-capita income, no carbon-intensive domestic electricity generation, and a low density of solar and wind parks. It would be worthwhile investigating how attitudes to solar parks compared to those of wind parks vary in countries with different characteristics, especially in countries in which citizens are more familiar with solar park installations.

Second, the research measured affective reactions to solar and wind energy on an abstract level, as well as reactions to pictures of energy installations. However, debates concerning the implementation of energy projects are often emotionally loaded, not only because of the nature of the energy infrastructure or its visual impact on the landscape, but also because of how project developers tackle the projects [84]. The current study investigated the acceptance of the public only and public opinion as part of the socio-political and community acceptance dimensions of the social acceptance triangle [27]. However, the concept of social acceptance of renewables is not only made up of mutually influencing processes at the three levels (community, socio-political, market) but also involves different actors (project developers, policymakers, consumers) who are active at different levels [29]. Further studies could investigate how emotions are elicited and influenced in line with the different levels of the social acceptance triangle [27]. This calls for longitudinal research that could potentially contribute to the call of Wolsink [29] to recognize social acceptance as a "bundle of dynamic processes instead of a set of actors' positions" (p.286).

Third, the aim of the study was not to predict individuals' actual intentions to act in favour or against a specific renewable energy project, but rather to compare attitudes towards solar and energy projects in a hypothetical setting. Investigating the attitude formation process early on, however, may be very important for forecasting local acceptance or voting intentions [85].

5.5. General conclusion and policy implications

Literature about the social acceptance of solar energy projects and specifically, large ones, is still scarce. Based on a representative survey in Switzerland (n = 601), the present results reveal, through a comparison of small- and large-size solar power installations with wind power installations, that the more positive affective reactions and attitudes enjoyed by solar energy significantly decrease as the size of the two technologies is similar. While earlier studies consistently found that people prefer solar over wind power, the present study adds some nuance to these results and hints at the fact that while large solar projects may appear promising, they may face similar challenges with acceptance of the population at the local level as wind energy does. In fact, the results show that solar energy is not always preferred to wind energy, and suggests that policymakers should not only rely on public opinions polls to form the basis for policy decisions, as these may lead to distorted perceptions about the public support for renewable energy projects. They should also remain aware of how these general preferences may change depending on the size of the energy projects and their visual impact on the landscape [45,87], and how these can affect people's values and elicited emotions (positive or negative), in addition to other factors, such as distributional and procedural justice [10,19,86] and place attachment [e.g. 20,49]. Policymakers should not shy away from addressing the fact that solar energy installations may not easily be scaled up from a community acceptance perspective, as is also the case with other technologies deployed on a large scale. For example, policymakers who hope they can avoid controversy by avoiding wind energy and prioritizing solar energy [88] may encounter similar problems with large-scale solar, or fail to meet the renewable energy targets defined in their national energy policies. Policymakers should thus consider social acceptance issues in general and, more specifically, acceptance at the local level, and think about how to diversify the energy mix to mitigate the risk of opposition. The current results also highlight the tension involved in choosing between the deployment of decentralized versus centralized renewable energy systems, as well as the related level of acceptance of the land use required for renewable energy systems weighed against alternative uses.

Findings further show that affect is an important factor underlying attitudes towards the implementation of solar parks, wind parks, and rooftop solar installations, and that affect elicited by pictures of energy infrastructures has a larger effect on attitudes in the case of large-scale installations. Further, the study highlights that affect elicited by solar energy in general is a weaker predictor of attitudes towards solar parks than towards rooftop solar installations. This supports earlier research [16] that found that individuals imagine small solar PV installations when thinking about solar energy at a general level, perhaps explaining the gap in public attitudes between wind and solar that is found in national surveys. Taken together, the findings show that as the deployment of renewables significantly increases to make Europe carbon neutral by 2050, policymakers and project developers may increasingly have to attend to the emotions elicited by energy infrastructure among the population during the different phases of the development of energy projects, and improve their communication. Attempts by policymakers or project developers to convey rational knowledge about the benefits of renewable energy projects may not lead to the intended behavioral outcomes unless emotional components are taken into account. Moreover, in relation to attitudes towards renewables, policymakers should be aware that there exists the risk that any technology, if deployed on a large scale, might face resistance, even if public attitudes are initially overwhelmingly positive.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Survey items

The following table lists survey items used to create the variables. Please note that the survey was fielded in German and French. The following is the English translation of the original survey.

Table A
Survey items used in the empirical analyses.

Variable	Survey item wording	Response scale
(1) Imagery/associations elicited by solar and wind energy at a general level	"What are the first thoughts or images that come to your mind when thinking of solar/wind energy?"	1 to 5 association listed
(2) Affective evaluations of the associations measured at a general level	"What are your feelings regarding the thought or images you provided about wind/solar energy?" *Please use the scale below to evaluate your feeling from very negative to very positive. Each thought or image needs to be evaluated separately.*	1-7 [very negative - very positive]
(3) Imagery/associations elicited by pictures of solar and wind energy installations	"What are the first thoughts or images that come to your mind when looking at this picture ?"	1 to 5 association listed
(4) Affective evaluations of the associations elicited by the pictures	"What are your feelings regarding the thought or images you provided about the picture?" *Please use the scale below to evaluate your feeling from very negative to very positive. Each thought or image needs to be evaluated separately.*	1-7 [very negative - very positive]
(5) Local concern	*Please indicate the extent to which you agree with the following statement:* "I would be concerned if a wind park/solar park/solar rooftop installation would be built in my neighborhood or other places, I feel close to".	1 (totally disagree) to 5 (totally agree) 6 (I do not know/no opinion)
(6) Gender	*n/a*	Male = 0; Female = 1
(7) Age group	*n/a*	1 = ≤ 29 2 = 30-45 3 = 45-59 4 = ≥ 59
(8) Language region	*n/a*	0 = German-speaking 1 = French-speaking
(9) Education	*n/a*	0 = no high school diploma 1 = high school diploma or higher
(10) Political orientation	*n/a*	1 = "Conservative" (SVP, FDP, CVP) 2 = "Center-progressive" (BDP, GLP) 3 = "Left-wing" (SP, GPS) 0 = no or other political party
(11) Place of residence	"Where do you currently live?"	1 = city centre 2 = suburbs 3 = agglomeration 4 = countryside 5 = no answer

Note: The questions on affective imagery (1–4) are based on Peters and Slovic [22]. Questions 1 and 5 were asked in a similar form in Cousse et al. [64].

Appendix B. Images presented to the respondents according to the treatment group

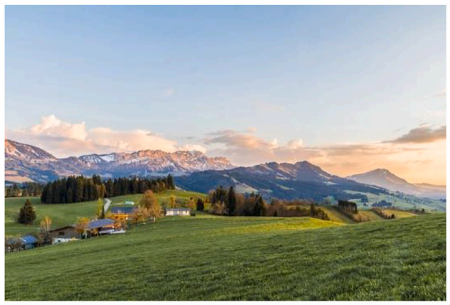
Treatment group	Scenario 1: Alpine	Scenario 2: Midlands
1. Solar park		
2. Solar rooftop		
3. Wind park		

Fig. B. Pictures of solar park, solar rooftop and wind park installations presented to the respondents based on their treatment group. (Felix Brönimann and author's figures, 2019)

Appendix C. Structure of the sample

Table C
Characteristics of the sample.

	"Wind park" group	"Solar park" group	"Solar rooftop" group	Total sample	Swiss population
	n = 201	n = 197	n = 203	n = 601	
Age					
15–29	17.90%	19.30%	18.20%	18.50%	23%
30–39	11.90%	17.30%	14.30%	14.50%	18%
40–49	19.90%	17.80%	22.20%	20.00%	19%
50–59	21.40%	19.80%	15.30%	18.80%	20%
60–74	28.90%	25.90%	30.00%	28.30%	20%
Gender					
♂	53.70%	49.70%	46.80%	50.10%	49%
♀	46.30%	50.30%	53.20%	49.90%	51%
Education					
low/medium	63.20%	61.90%	64.00%	63.10%	62%
high	36.80%	38.10%	36.00%	36.90%	38%
Political Orientation					
"Conservative" (SVP, FDP, CVP)	59.70%	53.30%	59.10%	57.40%	57%
"Center-progressive" (BDP, GLP)	9.50%	5.10%	9.40%	8.00%	9%
"Left-wing" (SP, GPS)	23.40%	32.50%	23.60%	26.50%	26%
Others/none	7.50%	9.10%	7.90%	8.20%	8%
Language region					
French-speaking	32.30%	38.60%	31.50%	34.10%	27%
German-speaking	67.70%	61.40%	68.50%	65.90%	73%

Note: Sources for the data about the Swiss population: Age [70], Gender [70], Political Attitude [70], and Language region [71].

Appendix D. Descriptive statistics

Table D
Descriptive statistics.

	Mean	S.D.	Min.	Max.	N
Average affect					
Affect towards solar (general)	5.74	1.83	1 (very negative)	7 (very positive)	601
Affect towards wind (general)	4.49	1.45	1 (very negative)	7 (very positive)	601
Affect towards solar park (S_1)	4.05	2.05	1 (very negative)	7 (very positive)	197
Affect towards solar park (S_2)	4.32	1.92	1 (very negative)	7 (very positive)	197
Affect towards wind park (S_1)	4.50	2.18	1 (very negative)	7 (very positive)	201
Affect towards wind park (S_2)	3.89	1.92	1 (very negative)	7 (very positive)	201
Affect towards solar rooftop (S_1)	6.43	1.04	1 (very negative)	7 (very positive)	203
Affect towards solar rooftop (S_2)	4.99	1.68	1 (very negative)	7 (very positive)	203
Socio-demographic variables					
Age	3.24	1.47	1 (<29)	4 (>59)	601
Gender	0.50	0.50	0 (male)	1 (female)	601
Education	0.37	0.48	0 (low/medium)	1 (high)	601
Language region	0.34	0.47	0 (German)	1 (French)	601
Political Orientation	1.66	0.90	1 ("Conservative")	3 ("Left-wing")	552
Income	1.95	0.84	1 (<6'000 CHF)	3 (>9'000 CHF)	512
Place of residence	2.97	1.17	1 (City center)	4 (Countryside)	596

Note: S.D. stands for "standard deviation".

Appendix E. Regression analyses including income

The analysis reported in Tables E1 and E2 is for assessing the sensitivity of the results to the inclusion of the income variable. As is well known for survey research, many respondents prefer not to answer income-related questions. This survey is not an exception with overall about 15% of respondents who preferred not to report their income levels. Acknowledging the potential source of biases in the coefficient estimates due to these dropped observations, this section nevertheless reports the regression results with the income variable (corresponding to Models 1a – 2c of the main specification in Tables 2–3). The estimated coefficients of income themselves turn out statistically not significant and the coefficients that were significant before are only slightly affected both in terms of their estimated magnitude and the confidence levels.

Table E.1
Multiple regression results concerning attitudes with general affect - controlling for income scale.

	Wind park (Model 1a')				Solar park (Model 1b')				Solar rooftop (Model 1c')			
	B	SE	β	p-values	B	SE	β	p-values	B	SE	β	p-values
General affect	−0.26	0.06	−0.32	0.00	−0.22	0.10	−0.19	0.02	−0.12	0.06	−0.17	0.04
Gender	−0.18	0.23	−0.06	0.44	0.19	0.26	0.06	0.46	−0.03	0.17	−0.02	0.84
Age	0.00	0.08	0.00	0.97	0.07	0.08	0.07	0.41	0.05	0.06	0.06	0.45
Language region	0.65	0.25	0.21	0.01	−0.17	0.26	−0.05	0.53	−0.51	0.19	−0.21	0.01
Education	−0.20	0.25	−0.07	0.42	0.13	0.26	0.04	0.61	−0.14	0.19	−0.06	0.45
Political orientation	−0.06	0.14	−0.04	0.64	−0.06	0.13	−0.04	0.65	−0.19	0.10	−0.15	0.06
Place of residence	0.02	0.10	0.01	0.86	0.19	0.11	0.14	0.09	−0.01	0.07	−0.01	0.91
Income	−0.17	0.142	−0.10	0.23	0.08	0.15	0.05	0.58	0.13	0.11	0.10	0.22
R^2 adjusted	0.13				0.02				0.09			
F	3.84*** (8, 144)				1.44 (8, 147)				2.85** (8, 151)			
Observations	153				156				160			

Table E.2
Multiple regression results concerning attitudes with the affect elicited by the energy installations - controlling for income scale.

	Wind park (Model 2a')				Solar park (Model 2b')				Solar rooftop (Model 2c')			
	B	SE	β	p-values	B	SE	β	p-values	B	SE	β	p-values
Affect – energy installation	−0.43	0.06	−0.50	0.00	−0.46	0.06	−0.55	0.00	−0.39	0.07	−0.42	0.00
Gender	−0.30	0.21	−0.10	0.15	0.37	0.22	0.12	0.10	−0.06	0.16	−0.03	0.69
Age	−0.08	0.08	−0.07	0.32	−0.04	0.08	−0.04	0.54	−0.03	0.06	−0.04	0.63
Language region	0.60	0.23	0.19	0.01	0.23	0.24	0.07	0.34	−0.43	0.17	−0.18	0.02
Education	−0.10	0.23	−0.03	0.68	0.07	0.23	0.02	0.77	−0.23	0.18	−0.10	0.20
Political orientation	−0.10	0.12	−0.06	0.42	−0.14	0.12	−0.09	0.21	−0.15	0.09	−0.11	0.11
Place of residence	0.02	0.09	0.0	0.79	0.10	0.10	0.07	0.32	0.05	0.07	0.06	0.45
Income	−0.17	0.13	−0.10	0.19	0.13	0.13	0.04	0.55	0.09	0.10	0.07	0.36
R^2 adjusted	0.27				0.25				0.21			
F	8.17 (8, 144)***				7.58 (8, 147)***				6.30 (8, 151)***			
Observations	153				156				160			

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